

Storm-Level Independent Verification of Operational Cloud Seeding: Evidence from Utah’s 2024–25 Winter Season

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Abstract

Operational winter cloud seeding programs claim precipitation increases of 3–15%, but the evaluations supporting these claims are produced by program operators using target/control methods that cannot separate seeding effects from *operator storm selection*: storms are seeded precisely when atmospheric conditions favor the target areas. This study performs, to the author’s knowledge, the first independent storm-level evaluation of an operational program built entirely from public data. We assemble a machine-readable record of Utah’s 2024–25 operations—181 generator sites, 203 seeded storm periods totaling 17,143 generator-hours across seven programs—together with per-storm 700 hPa winds from the HRRR archive and daily precipitation from 546 SNOTEL stations in seven states. Three research designs of increasing selection-robustness are applied. A naive seeded-versus-unseeded storm comparison yields an implausible +335% for the largest program, quantifying the selection artifact. Same-sector near-miss controls reduce the pooled estimate to +24% [+4, +50]. The most robust design—a wind-rotation test in which plume exposure is recomputed from each storm’s own wind field per generator—yields +1.5% [−9.6, +13.1] per seeded storm with a clean placebo on unseeded storms. One season therefore cannot distinguish the claimed 3–15% effect from zero, while per-storm effects larger than ~15–20% are excluded. The design’s power scales directly with additional seasons of operations records. All data and code are released.

Key words: Weather Modification, Cloud Seeding, Causal Inference, SNOTEL, HRRR, Program Evaluation

Introduction

Winter orographic cloud seeding disperses silver iodide (AgI) into supercooled liquid water within storms passing over mountain barriers, nucleating ice crystals that fall as additional snow. The physical mechanism is established: the SNOWIE field campaign traced seeding material through individual storms with airborne radar and observed unambiguous seeding signatures [1], and subsequent analysis quantified the snowfall attributable to individual seeding events [2]. What remains contested is the *operational* question: how much water do routine, ground-based seeding programs actually add, season over season?

The stakes of that question are growing. Utah operates the largest remotely-controlled seeding generator network in the United States, expanded in 2023 with \$12M in one-time funding and a \$5M annual budget following record-low levels in the Great Salt Lake [4]. The State’s published evaluations report target-area increases of 5–20% and an estimated 249,600 acre-feet of average annual augmented runoff at approximately \$1 per acre-foot [3]. Colorado River basin states co-fund seeding in Utah under multi-state agreements, meaning these estimates influence interstate water accounting.

These evaluations, however, share two structural weaknesses. First, they are produced by the contractor operating the programs. Second, and more fundamentally, they rest on historical target/control regressions whose treatment variable is not exogenous: *operators seed exactly those storms whose wind direction, moisture, and temperature favor the target ar-*

eas [3]. Suspension rules compound the problem — Utah’s criteria suspend seeding when snowpack exceeds 150–200% of average [3], making treatment anti-correlated with natural snowfall at seasonal scale and positively selected at storm scale. Any comparison of seeded versus unseeded periods, at any aggregation, inherits this selection unless the design explicitly breaks it.

This paper contributes three things. (1) A machine-readable reconstruction of one full season of Utah seeding operations from public documents: generator coordinates, per-storm seeding dates and generator-hours for all seven ground-based programs. (2) A sequence of three storm-level research designs applied to the same season of SNOTEL precipitation data, constructed so that each successive design removes a channel of selection bias; the progression of estimates (+335% → +24% → +1.5%) directly measures how much of a naive “seeding effect” is selection artifact. (3) A wind-rotation identification strategy, which to the author’s knowledge has not previously been applied to an operational program: because AgI plumes travel downwind, a station’s exposure changes storm-by-storm with the storm’s own wind direction, and a real seeding effect must rotate with it — something neither operator storm selection nor static orography can mimic.

Data

All inputs are public; acquisition and processing code is released with the paper.

Seeding operations records

The Utah Division of Water Resources publishes annual reports for each seeding program, prepared by the operator (North American Weather Consultants) [4]. From the eight 2024–25 reports we extract: (i) generator site tables — 181 sites with coordinates across the Central/Southern Utah + Tooele, Western Uintas, High Uintas, Northern Utah, East Shore, Six Creeks, and Cache Valley programs (Book Cliffs sites are published as town names only and are geocoded approximately); and (ii) storm-operations tables — 203 program-storm periods with dates, sites used, and generator-hours, totaling 17,143 hours across seven ground-based programs. Program treatment histories (adoption years, the 1983–87 statewide suspension) are taken from the operator’s 30-season retrospective [3].

Storm winds

For every storm period and detected precipitation event we compute the mean 700 hPa wind at each program’s generator network from the NOAA High-Resolution Rapid Refresh (HRRR) analysis archive on AWS Open Data [5], sampling 6-hourly analyses and extracting only the two relevant GRIB2 messages per file via byte-range requests. Storm-mean winds are vector-averaged. Consistent with program design assumptions, 95% of seeded storms have 700 hPa flow from the WSW–NW quadrant (median 264°).

Snowpack and precipitation observations

Daily accumulated-precipitation (PREC) series for all 546 SNOTEL stations in Utah, Colorado, Nevada, Wyoming, Idaho, Arizona, and New Mexico are drawn from the USDA NRCS Air-Water Database [6]. Daily increments are computed by first-differencing, with negative and physically implausible (>8 in/day) increments removed. A 45-year station-winter panel (1981–2026; 18,067 station-winters) is used for the seasonal-scale benchmark design.

Methods

Design 0: seasonal panel benchmark

As a baseline we estimate a staggered-adoption imputation estimator [7] on log November–March precipitation over 45 winters: station and winter fixed effects are fit on untreated observations (out-of-state controls, excluding ranges with their own seeding programs, plus pre-adoption years of Utah cohorts), and treatment effects are read off as anomalies of treated station-winters. Suspension years are excluded from fitting because they are endogenously wet.

Design 1: seeded vs. unseeded storms, far controls

Storm events are detected as contiguous days of regional precipitation. For each program, each event is labeled seeded/unseeded from the operations record; the outcome is the log ratio of mean storm-total precipitation at plume-exposed stations (≤ 40 km downwind of a generator) to that

at far control stations (> 90 km from all generators). The effect is the difference in mean log ratio between seeded and unseeded storms. This is essentially the operational-evaluation logic transposed to storm scale, and it is deliberately included as a measurement of selection bias.

Design 2: near-miss controls

Identical, except controls are “near-miss” stations 55–130 km from the program’s generators *in the same downwind sector*: they experience the same storm types that trigger seeding decisions but lie beyond AgI transport range. Storm-type selection therefore cancels to first order in the seeded-minus-unseeded contrast. Inference is by bootstrap over storm events. Two falsification tests are run: a placebo in which all seeded windows are shifted by +10 days, and a dose-response test correlating per-storm generator-hours with the per-storm effect.

Design 3: wind-rotation (per-generator plume geometry)

The most selection-robust design exploits within-station, across-storm variation in exposure. For each station within 70 km of a program’s generators and each detected event e , the station is *in-plume* if any generator lies 5–45 km upwind such that the bearing from generator to station is within $\pm 40^\circ$ of event e ’s downwind direction (from the HRRR storm-mean wind). We estimate

$$\log y_{se} = \alpha_s + \gamma_e + \delta \text{score}_{se} + \beta_1 P_{se} S_e + \beta_0 P_{se} (1 - S_e) + \varepsilon_{se},$$

where α_s, γ_e are station and program-event fixed effects, $\text{score}_{se} = \max_g \cos(\Delta\theta_{sg,e}) e^{-d_{sg}/40}$ is a smooth wind-alignment control absorbing orographic anisotropy that rotates with the wind, P_{se} is the in-plume indicator, and S_e indicates seeded events. β_1 is the seeding effect; β_0 is a built-in placebo that must be zero, since no AgI is released in unseeded events. Fixed effects are removed by iterated two-way demeaning; confidence intervals are by bootstrap over events. Selection cannot generate $\beta_1 > 0 = \beta_0$: operators choose *when* to seed, but they cannot rotate a precipitation enhancement around their generators in step with each storm’s wind.

Results and Analysis

Seasonal benchmark: the claimed effect is below the noise floor

The 45-year panel estimator yields an overall treatment effect of -1.4% (95% CI $[-6.0, +3.6]$) on winter precipitation across all Utah adoption cohorts, with a pre-adoption placebo of $+5.6\%$. Seasonal target/control analysis of station data therefore has a noise floor of roughly $\pm 6\%$ — wider than the lower half of the claimed 3–15% range. Published seasonal evaluations reporting significant 5–20% increases from comparable data should be interpreted with this in mind.

Table 1. Pooled storm-level estimates across designs, winter 2024–25. Effects are percentage changes in storm-total precipitation at exposed stations; 95% CIs by bootstrap over storm events.

Design	Effect	95% CI
1. Far controls (naive)	+69.0%	[−9.9, +220.8]
2. Near-miss controls	+24.2%	[+3.7, +49.7]
placebo: windows +10 days	+0.2%	[−16.7, +20.7]
3. Wind-rotation, per-generator	+1.5%	[−9.6, +13.1]
placebo: unseeded events	−8.4%	[−26.8, +10.3]

Source: author’s calculations from DWR operations reports, HRRR, and SNOTEL data; full per-program results in the released repository.

Selection bias is enormous at storm scale

Design 1 produces a pooled “effect” of +69%, and +335% for the Central/Southern program (Table 1) — an order of magnitude beyond physical plausibility [2] and a direct quantification of operator storm selection: seeded storms are storms chosen for favoring the targets.

Near-miss controls: directionally positive, still inflated

Design 2 collapses the pooled estimate to +24.2% [+3.7, +49.7] over 121 program-storm observations. The +10-day placebo returns +0.2% [−16.7, +20.7], confirming the pipeline produces no mechanical effect; per-storm generator-hours correlate positively but insignificantly with the per-storm effect (Spearman $\rho = 0.13$, $p = 0.26$, $n = 75$). The estimate nevertheless remains above the physically plausible range, consistent with residual selection: orographic enhancement on strong-flow seedable storms is naturally larger at the barrier than 55–130 km downwind, and near-miss controls only partially absorb this.

Wind-rotation: no resolvable effect at one season

Design 3 yields $\beta_1 = +1.5\%$ [−9.6, +13.1] for in-plume exposure during seeded events, with the unseeded-event placebo at $\beta_0 = -8.4\%$ [−26.8, +10.3] — statistically zero, as required. (An intermediate specification using program-centroid rather than per-generator geometry produced an unseeded-event artifact of +12%, illustrating that plume geometry matters: several Utah generator networks form north–south lines exceeding 100 km, which a centroid represents poorly.)

The point estimate sits exactly where the physical literature suggests a true operational effect should — small and positive — but one season of data cannot distinguish it from zero. Conversely, per-storm effects larger than approximately 15–20% are excluded at conventional confidence levels.

Power and the path to resolution

The binding constraint is seasons, not stations. Winter 2024–25 supplies 26 usable events and 121 program-event observations, giving Design 3 a CI half-width of ~ 11 percentage points. Under simple $1/\sqrt{n}$ scaling, four to nine additional

seasons of operations records — which the Division of Water Resources maintains and provides on request — would narrow the CI to ± 4 –5%, sufficient to confirm or refute the 3–15% operational claims. The marginal cost of extending the analysis is transcription of the operators’ storm tables; all other inputs (HRRR, SNOTEL) extend automatically.

Discussion

Three conclusions follow. First, the headline numbers used to justify operational seeding budgets are produced by methods whose selection bias, at storm scale, we measure at more than an order of magnitude larger than any plausible true effect; seasonal versions of those methods have noise floors wider than the effects claimed. Second, the most selection-robust design available finds a small positive point estimate fully consistent with both the gold-standard physical evidence [2] and with zero — after one season. The correct current statement about Utah’s program is therefore agnostic: the claimed effect is neither confirmed nor refuted, and effects large enough to matter for interstate water accounting (≥ 15 –20% per storm) are ruled out. Third, extra-area (“downwind”) effects on neighboring regions — a recurring political concern — are necessarily smaller than target-area effects and hence well below ground-based detectability at current program intensities.

Limitations

Wind sectors use each storm’s mean 700 hPa flow at the network; hourly wind shifts within storms are not modeled. SNOTEL gauges undercatch snowfall in wind, which attenuates but does not bias the estimates given fixed effects. Book Cliffs generator locations are approximate. The seeded/unseeded event labels inherit any errors in the operators’ storm tables. Finally, the event catalog is built from precipitation, so completely dry seeded periods are unobservable; this biases against detecting effects only if seeding generates precipitation where none would otherwise occur, which the physical literature does not support at meaningful scale.

Data and Code Availability

All datasets constructed for this study (generator coordinates, storm operations record, storm winds, event-level results) and all analysis code are released at the accompanying repository. Every input derives from public sources: Utah DWR seasonal reports, NOAA HRRR (AWS Open Data), and USDA NRCS SNOTEL.

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